

Measuring Diversity and Segregation with Convex Functions

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The study of quantitative measures for residential segregation extends far enough back that a review article in 1955 could express frustration and disappointment with the chaotic state of the methodology. Commenting then on the measures extant at the time, Duncan and Duncan [DD55] wrote:

1. *The proposed indexes of segregation have a number of hitherto unnoticed interrelationships which can be mathematically demonstrated.*
2. *Some of them have mathematical properties of which their proponents were unaware, and which lead to difficulties of interpretation.*
3. *As a consequence, the status of the empirical work already done with segregation indices is questionable, and their validity for further research is undermined.*

Over 70 years later, hints of this chaos remain. This author is aware of two instances in the last ten years of a segregation measure being published as novel despite already existing in the literature under a different name.

This article is not a systematic review of the literature, but rather a guided tour through one particular perspective on the problem of measuring segregation. This perspective begins with a general class of measures of diversity and segregation which

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Broad motivation for the article

- Folks in the math community are turning to quantitative measures of segregation [DMW23, TBK⁺25, HSS25].

- There’s actually a pretty interesting and rich literature in quantitative sociology on this set of topics [Rea, RF02], going back all the way at least to [Sim49].
- MAUP is a thing [Ope84], and we’d like measures that behave in predictable ways as we change analytical scale.
- If you start engaging with these ideas systematically and mathematically, then you come naturally to convexity and Bregman divergences, which are what I am here to sell you today.

Figure 1 is a simple checkerboard pattern. What if we actually oriented the entire piece around the checkerboard¹:

Preliminaries

To make these ideas concrete, we formalize the idea of a *population* and measurements which reflect the diversity or segregation of that population. Our framework for defining populations is intentionally discrete, in recognition of the fact that all Census-type data sets provide demographic information in discrete areal units such as counties, tracts, or blockgroups.

Definition 1. For $\ell \in \mathbb{N}$, the *probability simplex* Δ_ℓ is the set

$$\Delta_\ell = \left\{ \mathbf{y} \in \mathbb{R}^\ell : \sum_{j=1}^{\ell} y_j = 1, y_j \geq 0 \text{ for all } j \in [\ell] \right\}. \quad (1)$$

¹The problem of constructing segregation measures that distinguish between panels (c) and (d) in Figure 1 is often called the “checkerboard problem.”

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of probability vectors on ℓ elements.²

Definition 2. A population $\mathcal{P}$ with n locations in d spatial dimensions and ℓ demographic categories is a triple $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu)$, where:

- $\mathcal{X}$ is a finite metric space with n elements labeled $x_i, i = 1, \dots, n$.
- $\mathbf{Y} \in \mathbb{R}^{n \times \ell}$ is a row-stochastic matrix: for each row i , $\mathbf{y}_i \in \Delta_\ell$.
- $\mu \in \Delta_n$.

We denote the set of all populations with n locations, d dimensions, and ℓ demographic categories as $\mathcal{P}_{nd\ell}$.

In this definition, the matrix $\mathcal{X}$ represents space: element x_i of $\mathcal{X}$ can be viewed as giving the location of a population unit i . The matrix $\mathbf{Y}$ represents the demographic distribution of the population according to some attribute such as gender, race, or education level: the i th row $\mathbf{y}_i$ of $\mathbf{Y}$ gives the proportion of individuals in each demographic category at location i . Finally, the probability vector μ represents the population density across locations; the proportion of individuals living in location i is μ_i . The probability that a single individual, chosen uniformly at random from the population, lives in location i and is in demographic category j is $\mu_i y_{ij}$. The mean population vector $\bar{\mathbf{y}} = \sum_{i=1}^n \mu_i \mathbf{y}_i$ has entries giving the marginal probabilities that an individual chosen uniformly at random belongs to each demographic category.

Definition 3. A *population measurement* is a function $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$.³

An example of a population measurement is $f(\mathcal{P}) = \sum_{i=1}^n \mu_i^2$, which measures the degree to which the population is concentrated in space. For fixed n , the measure f is maximized (with value $f(\mathcal{P}) = 1$) when the entire population resides in a single location x_i and minimized (with value $f(\mathcal{P}) = \frac{1}{n}$) when the population is uniformly distributed across all locations.

²Although it is sometimes common to refer to Δ_ℓ as an $(\ell - 1)$ -simplex (this being its manifold dimension), we write Δ_ℓ to emphasize its role as the set of probability distributions over ℓ categories.

³The function η could reasonably be described as a *population measure*, which is an entirely disjoint concept from a population measurement. We will avoid the phrase “population measure” and emphasize “measurement” whenever possible.

Diversity

Segregation

Spatial Scale

Extensions

Figures we want

- Checkerboards for motivating the kinds of structures we want to distinguish between.
- Illustration of hierarchical units of regionalization on a map.
- Visualization of local information on the chessboard and / or on one or more real maps.
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- What properties to have? Pielou criteria [Pie77], various axioms from [RF02], etc.

This document can serve as a template for your article. Here are a few things to keep in mind. Articles should be understandable by 2nd year math grad students, and should be in the style of a good colloquium talk where the first $\frac{3}{4}$ is non-technical, and the last $\frac{1}{4}$ may be technical (but certainly does not have to be). Your article cannot have an abstract or an author bio. All figures need to have captions. You must adhere to the maximum length and number of references given in the SubmissionGuidelines file.

[Cho25] [Cho17]
[BMDG05]

References

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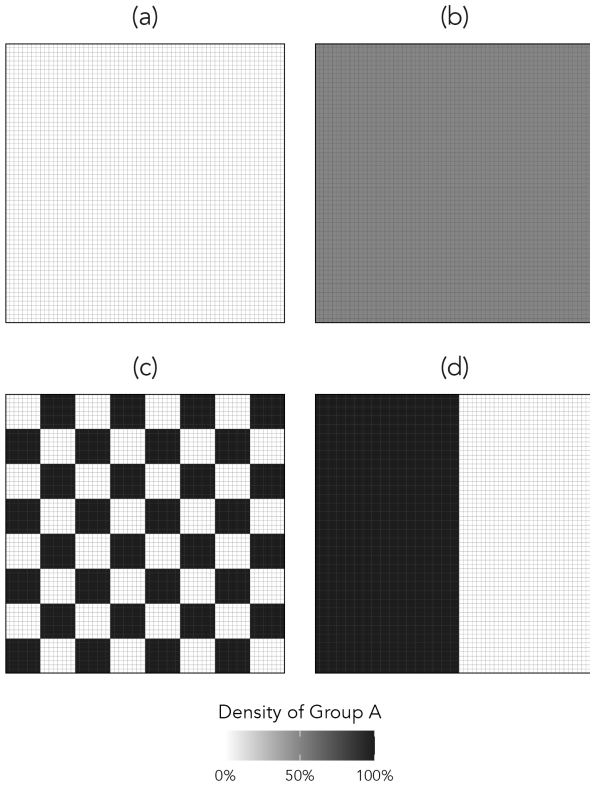


Figure 1: A simple checkerboard pattern. Each of the four panels is composed of 64 cells. Panel (c) is further arranged into 8 blocks of 8 cells each.

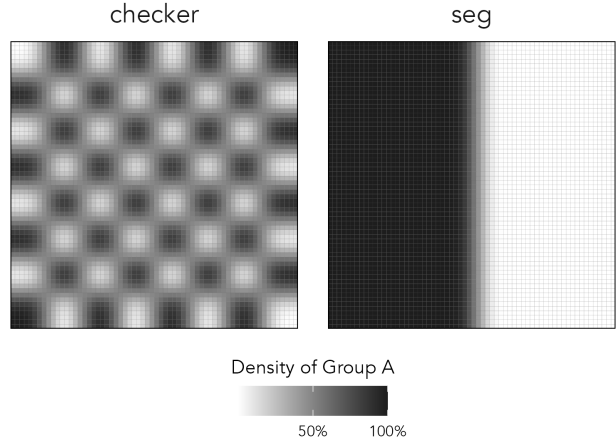


Figure 2: The smoothing bandwidth is 2 pixels.

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